radii), so ultimately $|C_BB|+|BZ|>|ZA|+|AC_B|$ $\Rightarrow|BZ|>|ZA|$.
Therefore, the minimum distance is the length of segment from Z to S_B's sphere, along the direction connecting its centroid to Z.

Likewise, we can add a threshold argument to the search function, which become the initial pruning distance (instead of passing infinity as the default value for `nnDist`), to also support search in spherical regions. Since these implementations can be trivially obtained, referring to chapter 9 for guidance, we leave them to the readers (for a hint, check the implementation on the book's repo).

10.4.2 Region Search

Region search will be similar to what we have described for k-d trees: the only difference being how we need to compute the intersection between each node and the search region, besides the structural change due to the fact that here points are only stored in leaves.

Listing 10.20 The pointsWithinRegion method for Ss-tree

```
function pointsWithinRegion(node, region)
                  #1
  points ←
[]
    #2
  if node.leaf then
                                                    #3
```